

SLIDES

Lesson 1 — The workbook as a test lab

AI agents on the workbook

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The workbook as a test lab

AI agents that flash, measure and test on their own



THE AI AGENT CAN:

- Flash firmware
- Watch the serial log
- Run tests
- Validate the results
- Fix problems in the code if needed

→ Workbench is an automated test lab

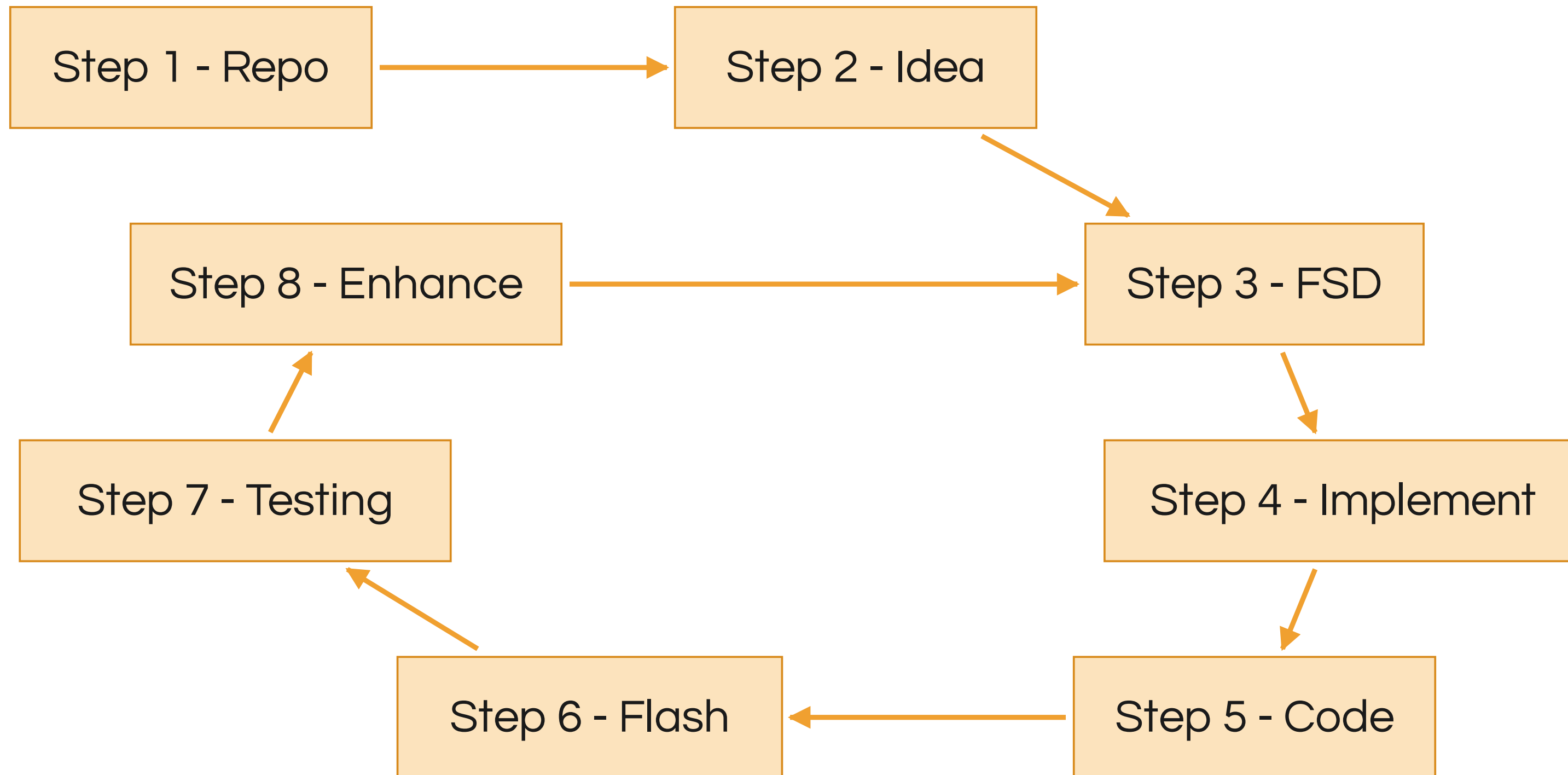


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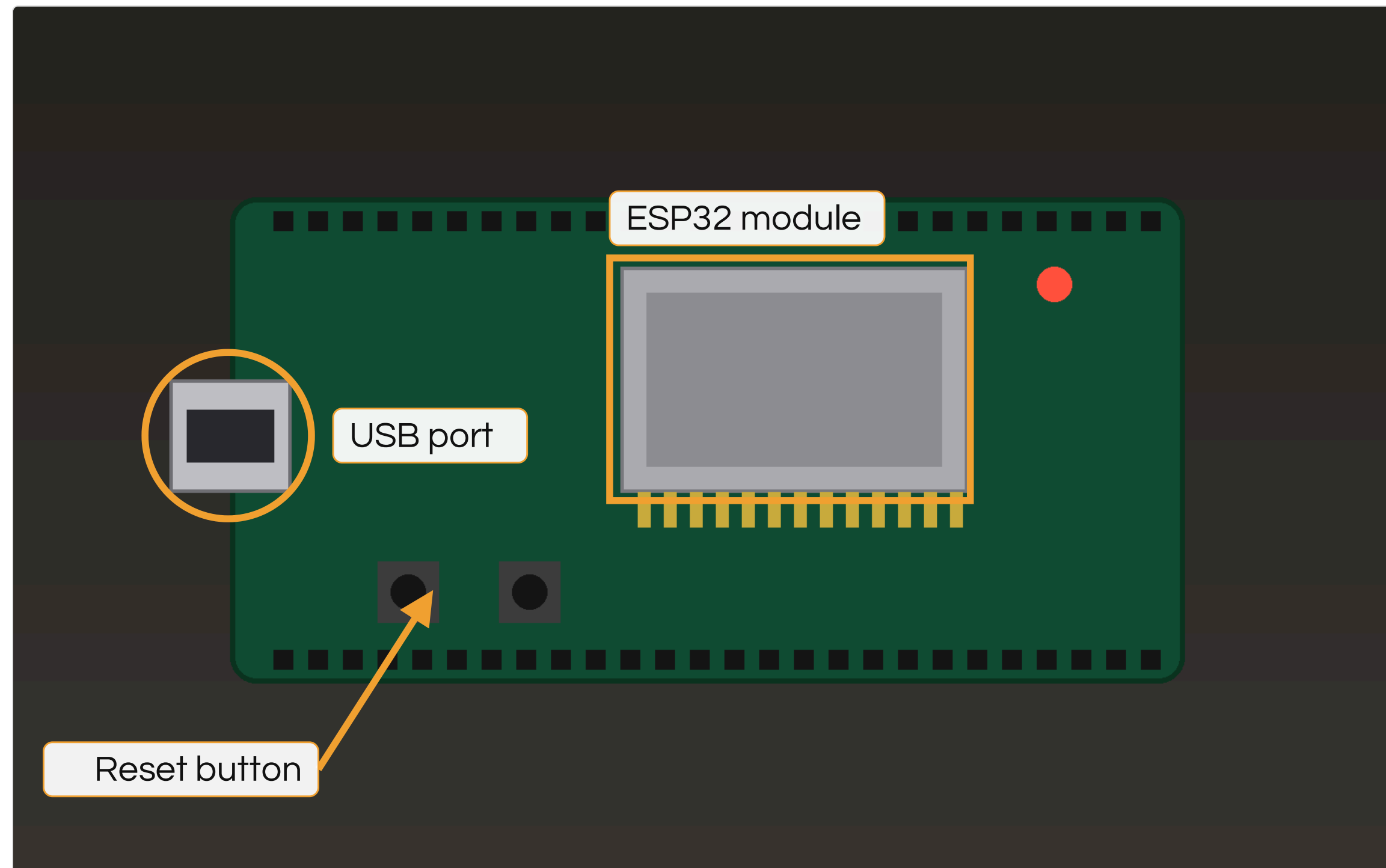
Let the machine do the boring work, and
keep the overview yourself.



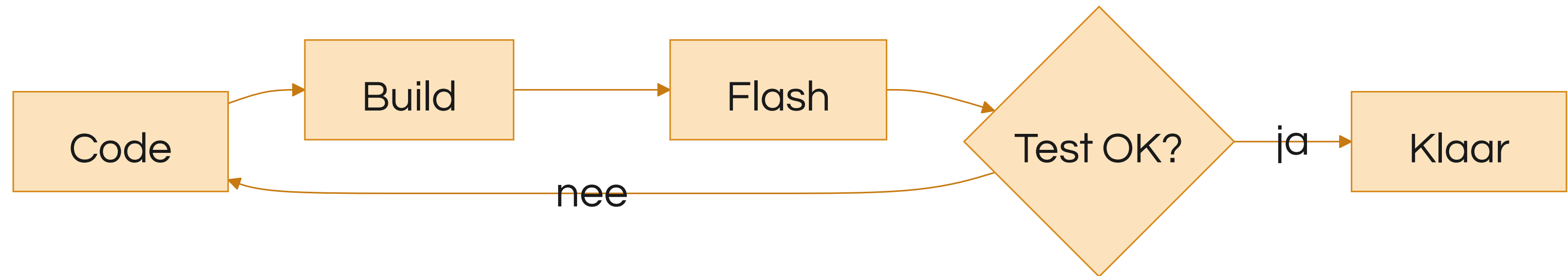
Ishikawa Circle



THE BOARD



THE TEST LOOP



A TEST IN PYTHON

```
import serial
port = serial.Serial("/dev/ttyUSB0", 115200)

def wait_for(text, timeout=10):
    line = port.read_until(text.encode(), timeout)
    return text in line.decode()

assert wait_for("BOOT OK")
```



BAUD RATE AND TIME

$$t_{byte} = \frac{10}{\text{baud rate}}$$

At 115200 baud, one byte takes about $87 \mu s$.



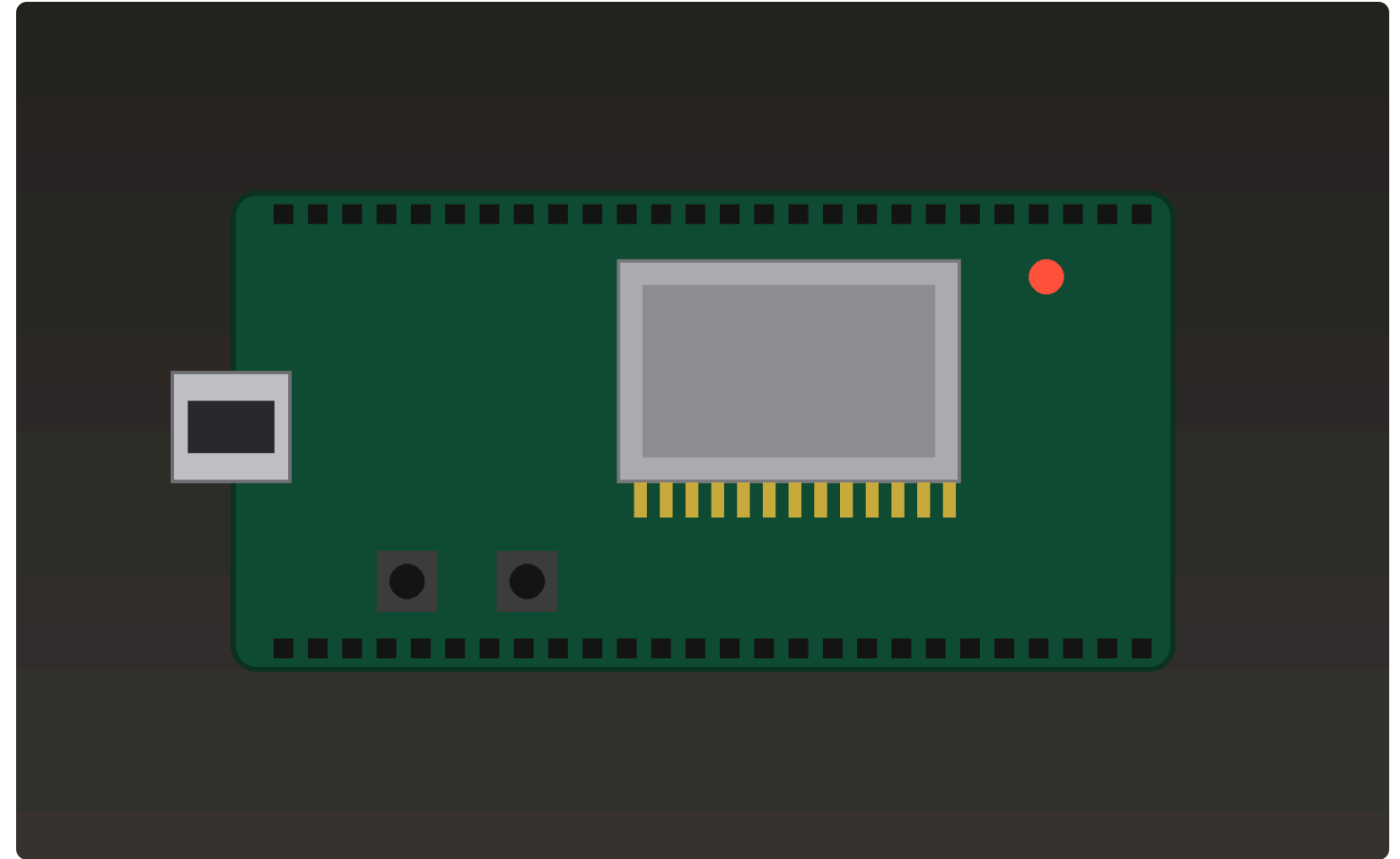
MANUAL OR WITH AN AGENT?

Task	Manual	With agent
Flashing	1 min	automatic
Reading logs	eyes on screen	continuous
Testing	when you remember	every build



SETUP

- Laptop with agent
- USB hub
- Test board



Thank you!

Questions? Comments? Let me know.